Supplementary Appendix 3

NCPE

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1 Importance sampling: model diagnostics, parameter distributions, and survival estimates

The tables and figures in this section show survival curve parameter estimates, covariance structures, importance sampling diagnostics, and (probabilistic) survival curve predictions, obtained from trial data only estimated using maximum likelihood estimation (MLE), and combining trial data with external information using the importance sampling algorithm described in the methods section of the main article and in Appendix 1.

1.1 Comparisons of parameter estimates with and without external information

The below tables compare parameter estimates and covariance matrices obtained from trial data only with those obtained combining trial data with external information, for each of the survival models under consideration.

Table 1: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Exponential distribution.

	x		x
rate	-4.562874	rate	-4.467088

Table 2: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Exponential distribution.

	rate		rate
rate	0.0169492	rate	0.0138444

Table 3: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Weibull distribution.

	x		x
shape	0.2028128	shape	0.2199019
scale	4.2127617	scale	4.1750757

Table 4: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Weibull distribution.

	shape	scale		shape	scale
shape	0.0147844	-0.0228526	shape	0.0094519	-0.0118402
scale	-0.0228526	0.0466214	scale	-0.0118402	0.0235770

Table 5: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Gompertz distribution.

	x		x
shape	0.0389625	shape	0.0197027
rate	-4.8682858	rate	-4.7220930

Table 6: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Gompertz distribution.

	shape	rate		shape	rate
shape	0.0006747	-0.0056127	shape	0.0000970	-0.0013281
rate	-0.0056127	0.0636371	rate	-0.0013281	0.0316261

Table 7: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Log-normal distribution.

	x		x
meanlog	4.5379239	meanlog	4.2299230
sdlog	0.5475716	sdlog	0.4404943

Table 8: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Log-normal distribution.

	meanlog	sdlog		meanlog	sdlog
meanlog	0.0696811	0.0247171	meanlog	0.0545435	0.0201378
sdlog	0.0247171	0.0114622	sdlog	0.0201378	0.0100655

Table 9: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Log-logistic distribution.

	x		x
shape	0.2450486	shape	0.3103378
scale	4.0866536	scale	3.9558418

Table 10: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Log-logistic distribution.

	shape	scale
shape	0.0144929	-0.0213003
scale	-0.0213003	0.0434085

	shape	scale
shape	0.0111243	-0.0139387
scale	-0.0139387	0.0276612

Table 11: Parameter estimates obtained using trial data only (left) and combining trial data with external information (right): Gen. Gamma distribution.

	x
mu	4.1703317
sigma	-0.3599072
Q	1.2031485

	x
mu	4.1857610
sigma	-0.3818327
Q	1.2508022

1.2 Effective sample size plots

The below plots show the estimated effective sample size (ESS), corresponding to 5000 (weighted) parameter samples, plotted as a function of the tempering parameter alpha, for the survival distributions under consideration. The vertical line shows the optimal value of alpha identified by the algorithm. Larger ESS values are better.

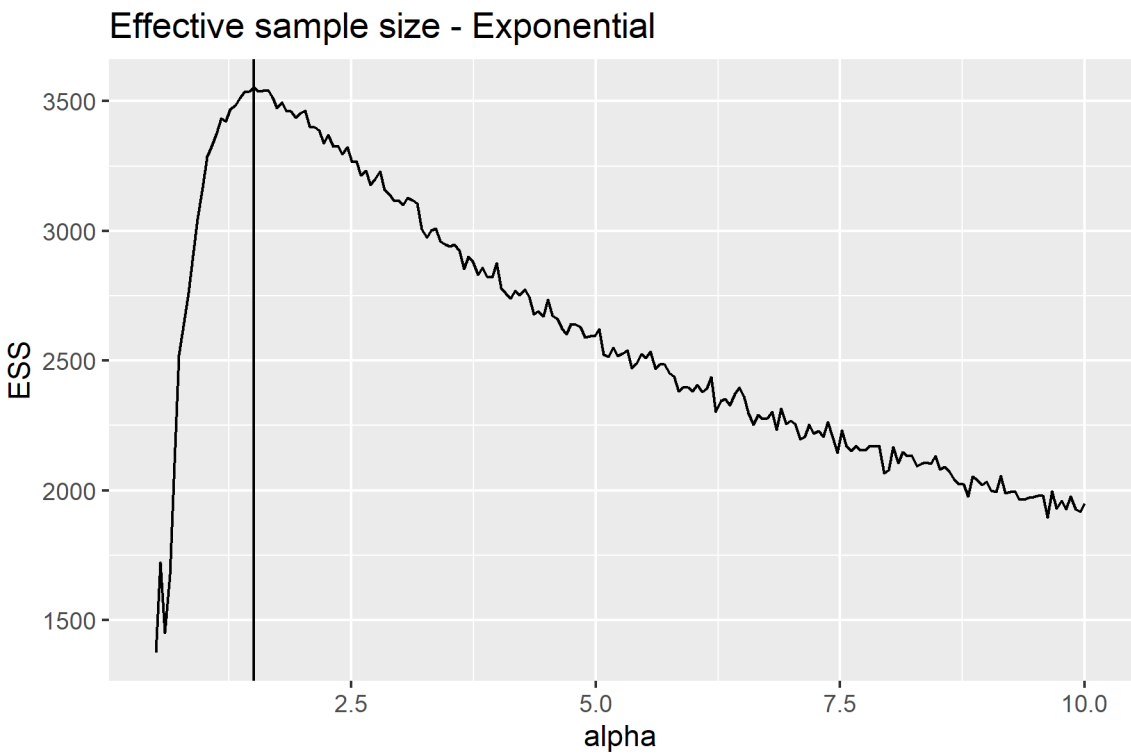
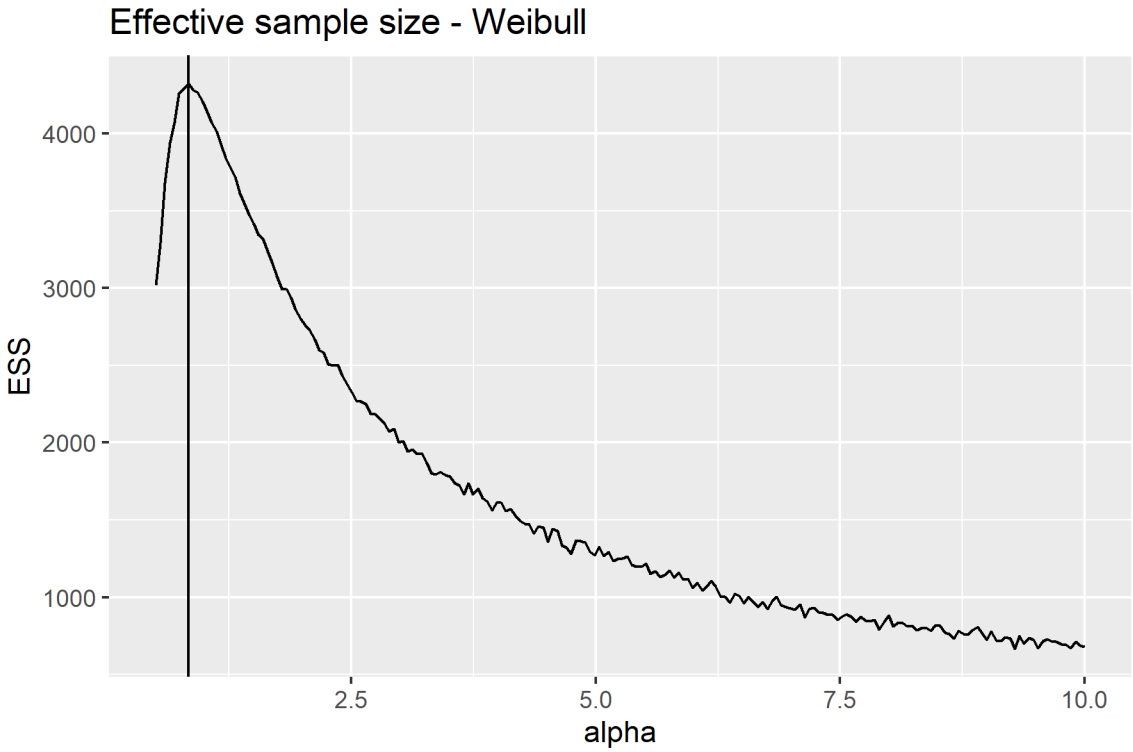
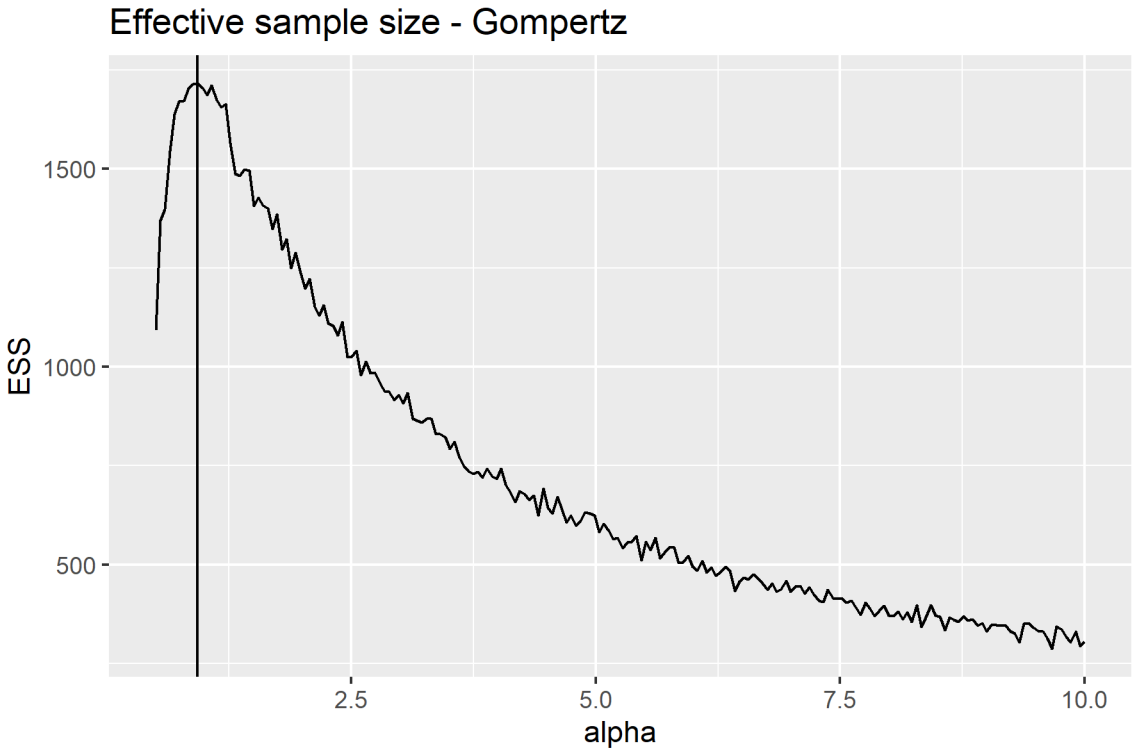


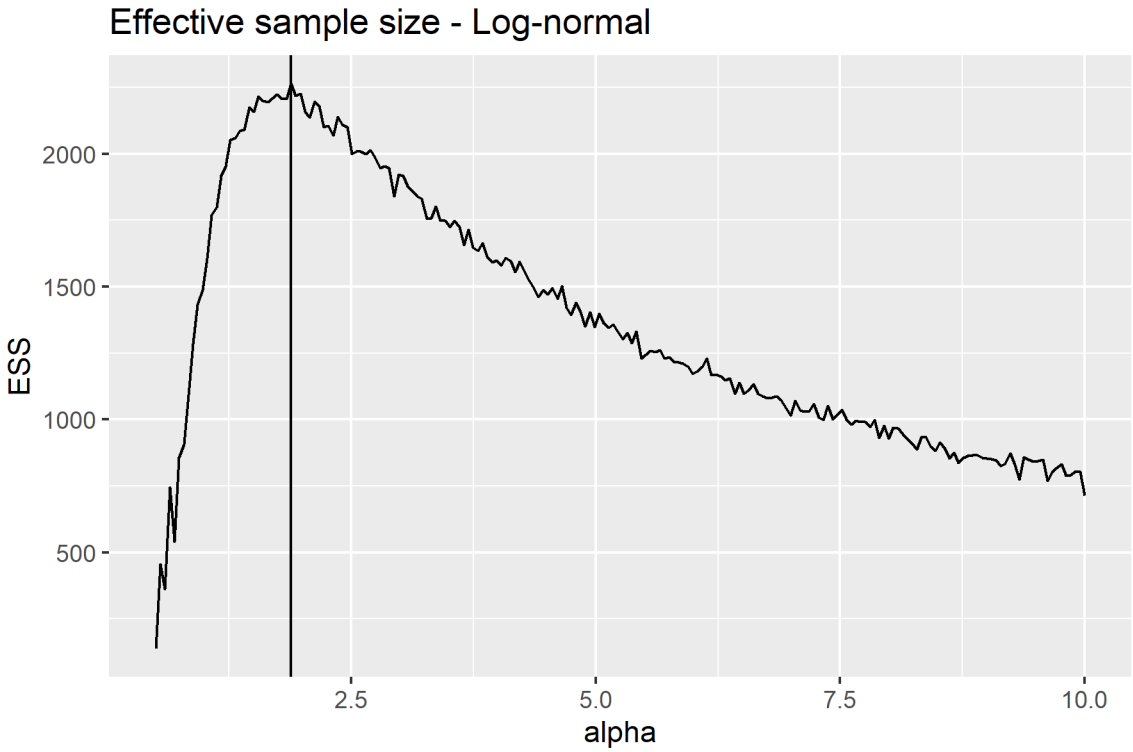
Table 12: Covariance matrices obtained using trial data only (left) and combining trial data with external information (right): Gen. Gamma distribution.

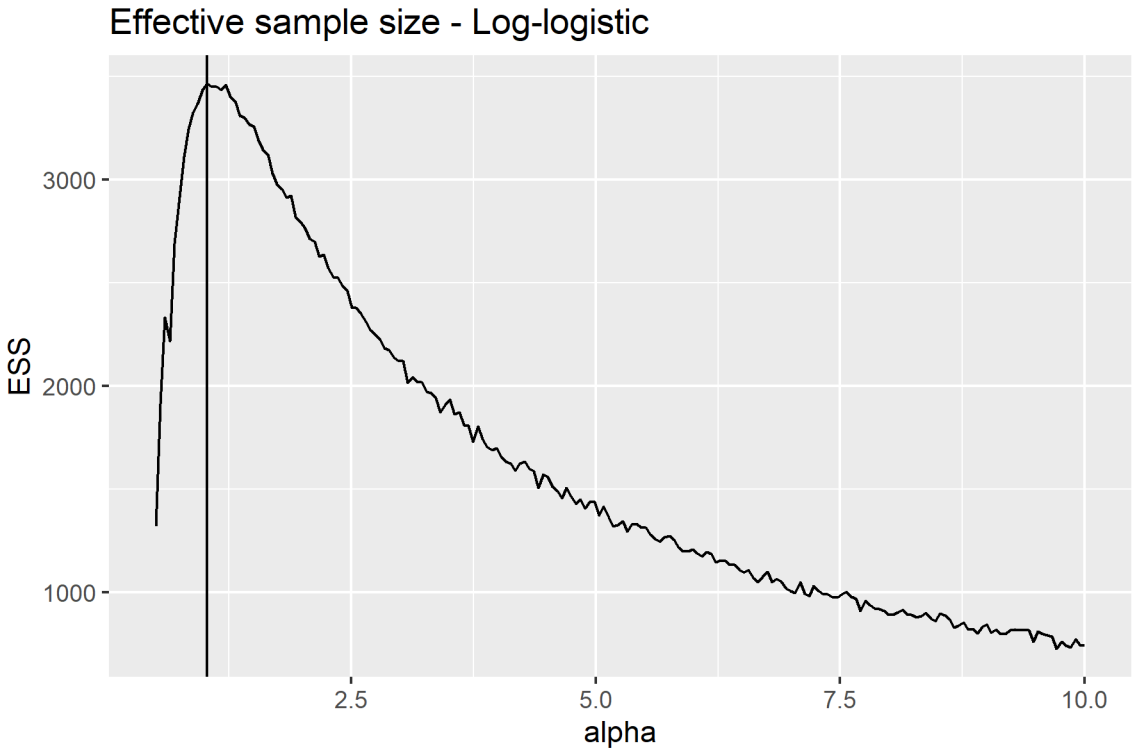
	mu	sigma	Q
mu	0.1026807	0.2899736	-0.3589428
sigma	0.2899736	1.2303463	-1.6290913
Q	-0.3589428	-1.6290913	2.1837103

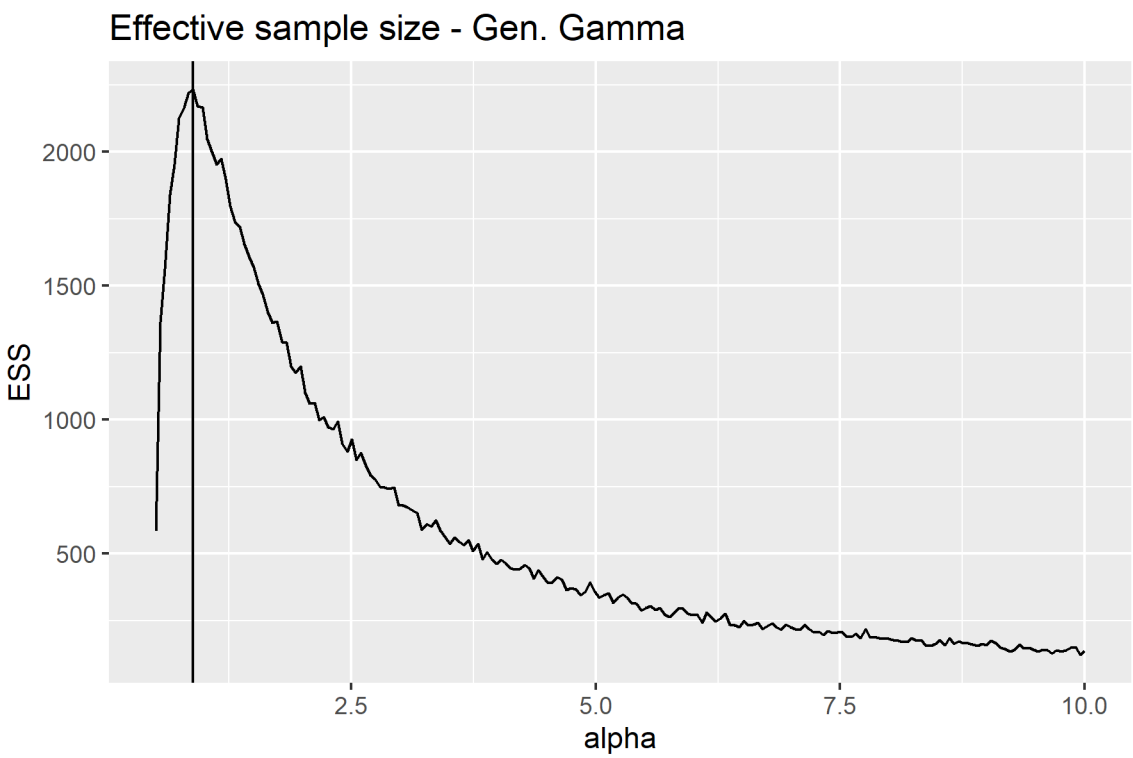
	mu	sigma	Q
mu	0.0219135	0.0031467	0.0125013
sigma	0.0031467	0.2238160	-0.3291366
Q	0.0125013	-0.3291366	0.5057176


Effective sample size - Weibull
ESS
4000
3000
2000
1000
2.5
5.0
7.5
10.0
alpha


Effective sample size - Gompertz
ESS
1500
1000
500
2.5
5.0
7.5
10.0
alpha

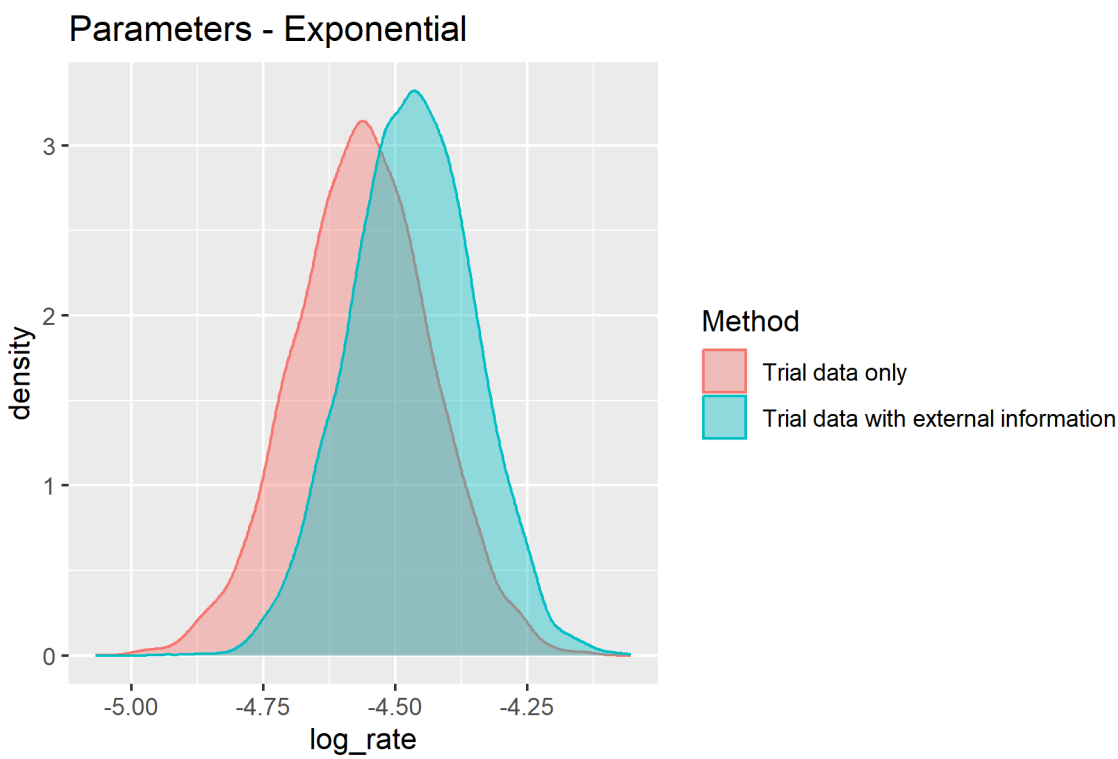

Effective sample size - Log-normal
ESS
2000
1500
1000
500
2.5
5.0
7.5
10.0
alpha

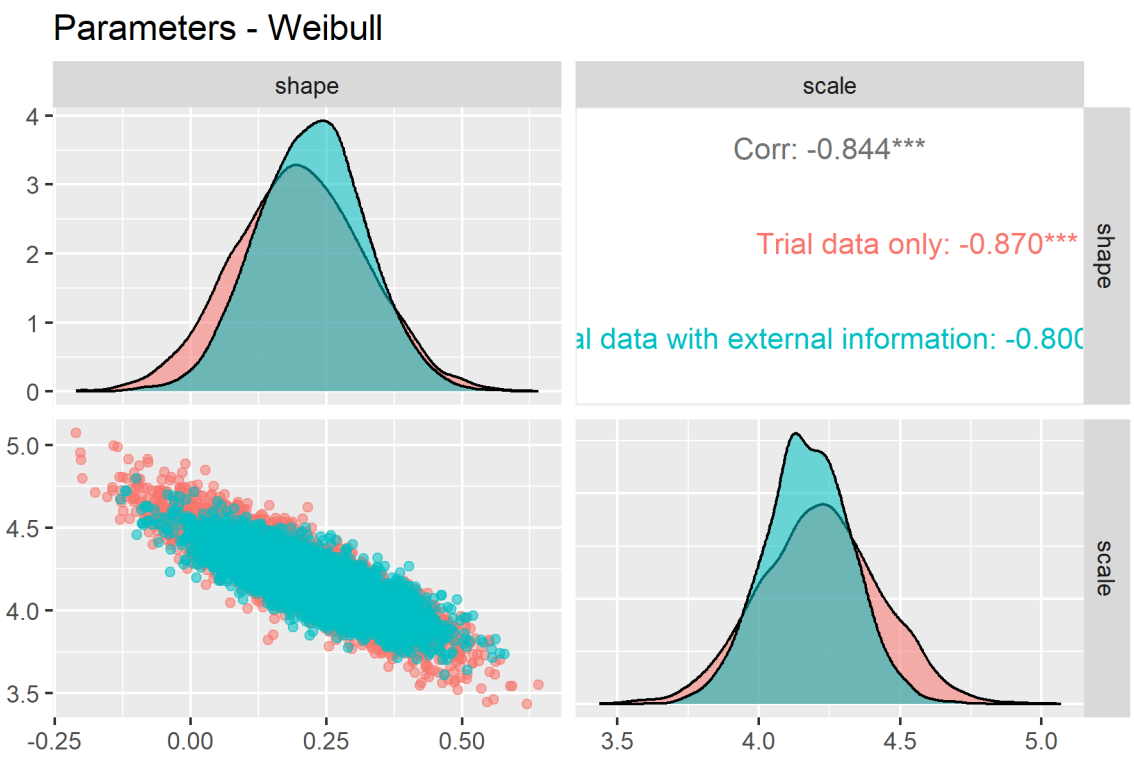

Effective sample size - Log-logistic
ESS
3000
2000
1000
2.5
5.0
7.5
10.0
alpha

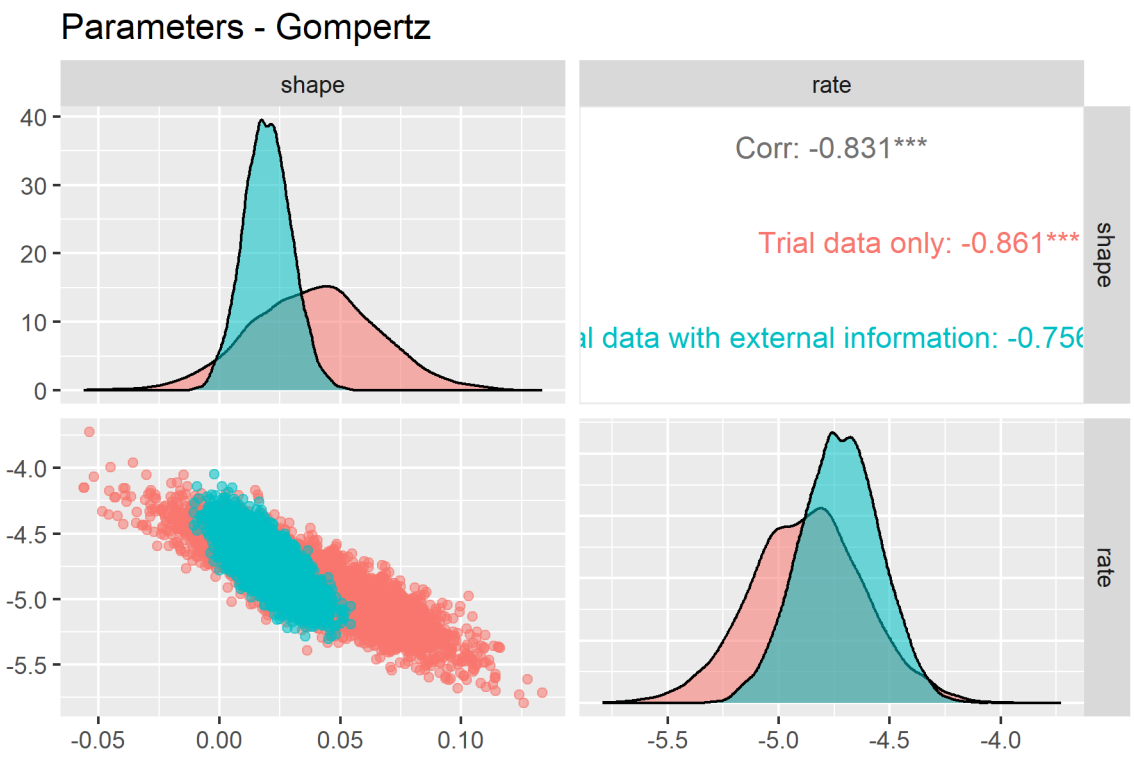


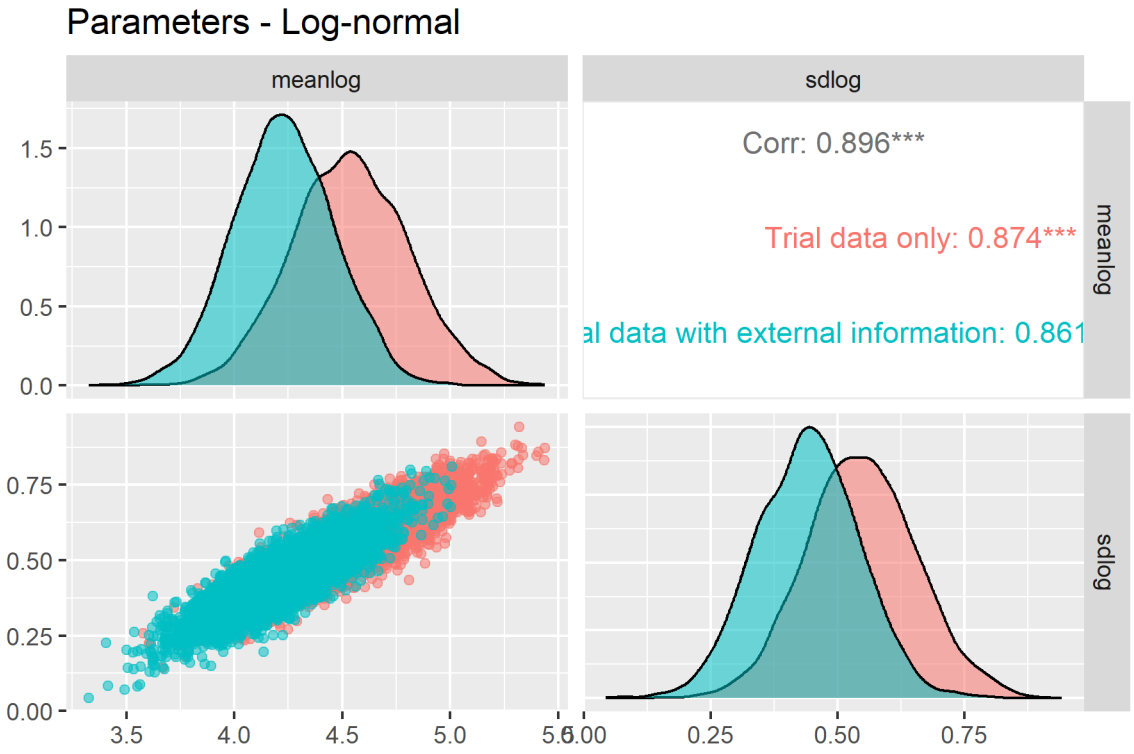
1.3 Distribution comparison plots

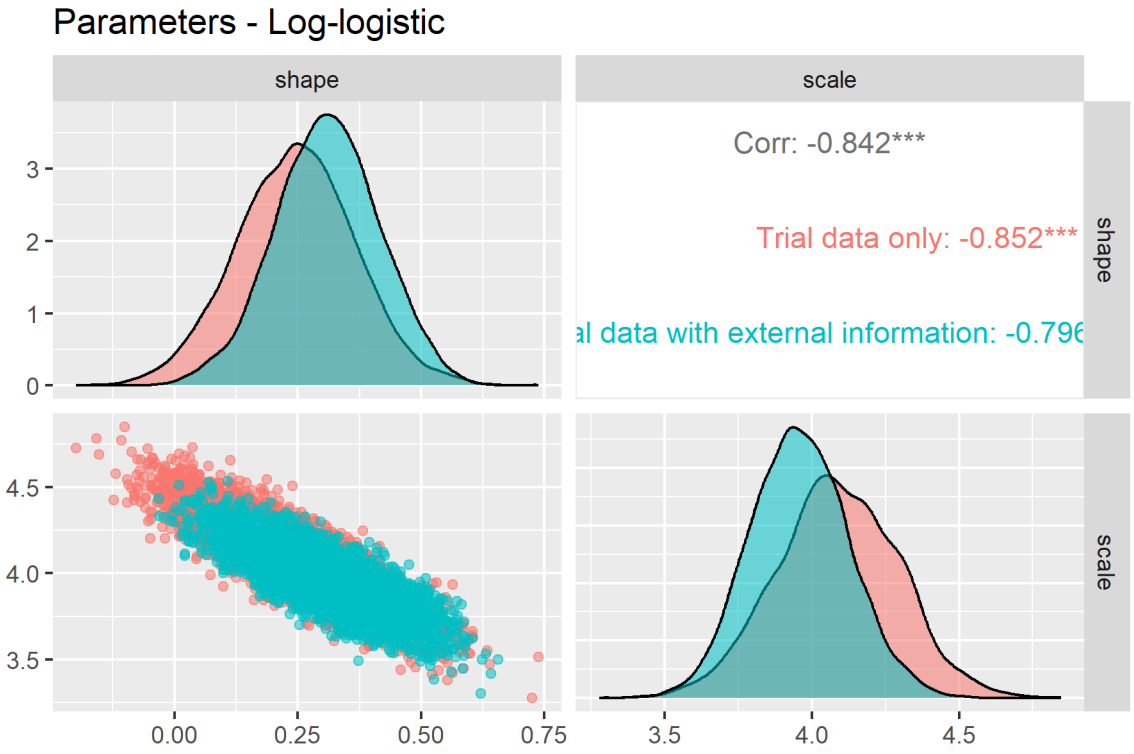
The below plots show parameter distributions and scatter plot estimates obtained using trial data only and by combining trial data with external information for the survival distributions under consideration.

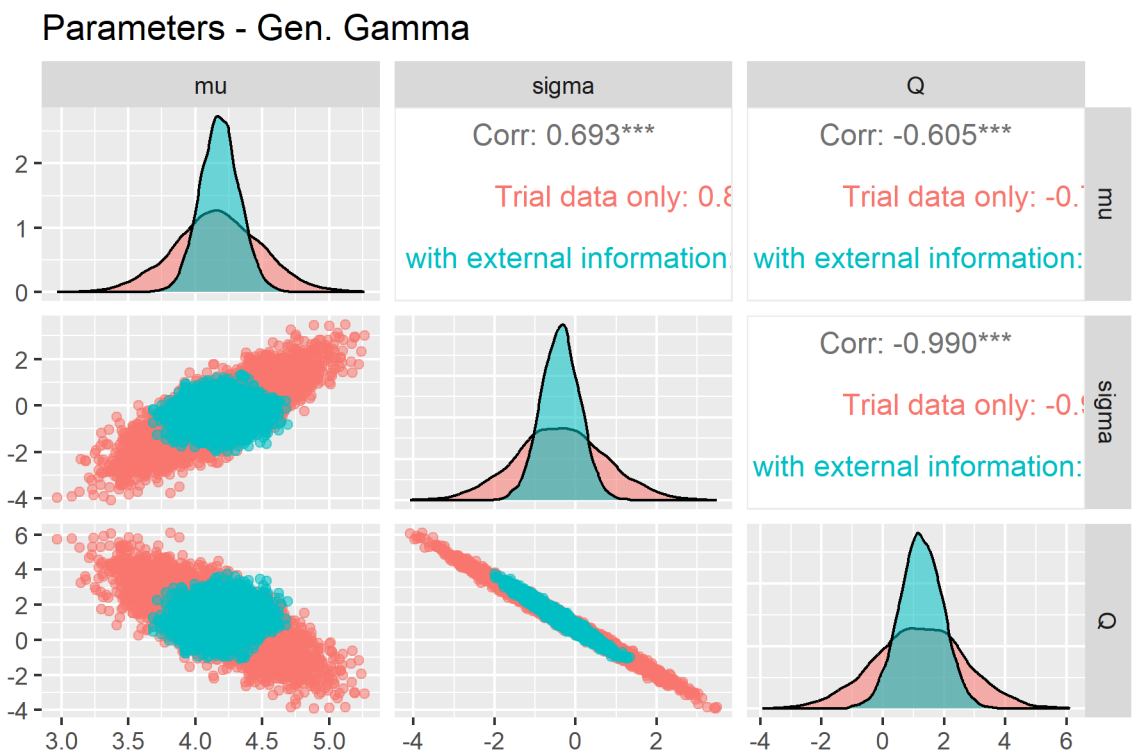



Parameters - Weibull
shape
scale
Corr: -0.844***
Trial data only: -0.870***
al data with external information: -0.800
shape
scale
4
3
2
1
0
5.0
4.5
4.0
3.5
-0.25
0.00
0.25
0.50
3.5
4.0
4.5
5.0


Parameters - Gompertz
shape
rate
Corr: -0.831***
Trial data only: -0.861***
al data with external information: -0.756
shape
rate
40
30
20
10
0
-4.0
-4.5
-5.0
-5.5
-0.05
0.00
0.05
0.10
-5.5
-5.0
-4.5
-4.0

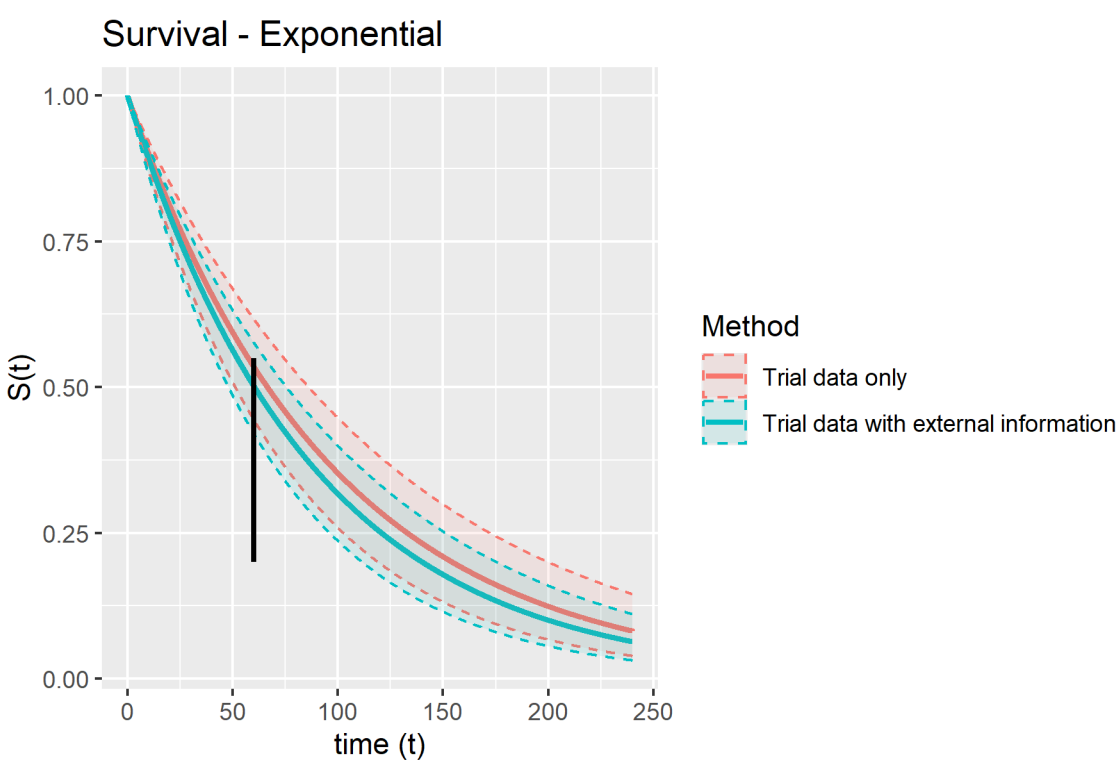

Parameters - Log-normal
meanlog
sdlog
Corr: 0.896***
Trial data only: 0.874***
al data with external information: 0.861
meanlog
sdlog
1.5
1.0
0.5
0.0
0.75
0.50
0.25
0.00
3.5
4.0
4.5
5.0
0.00
0.25
0.50
0.75

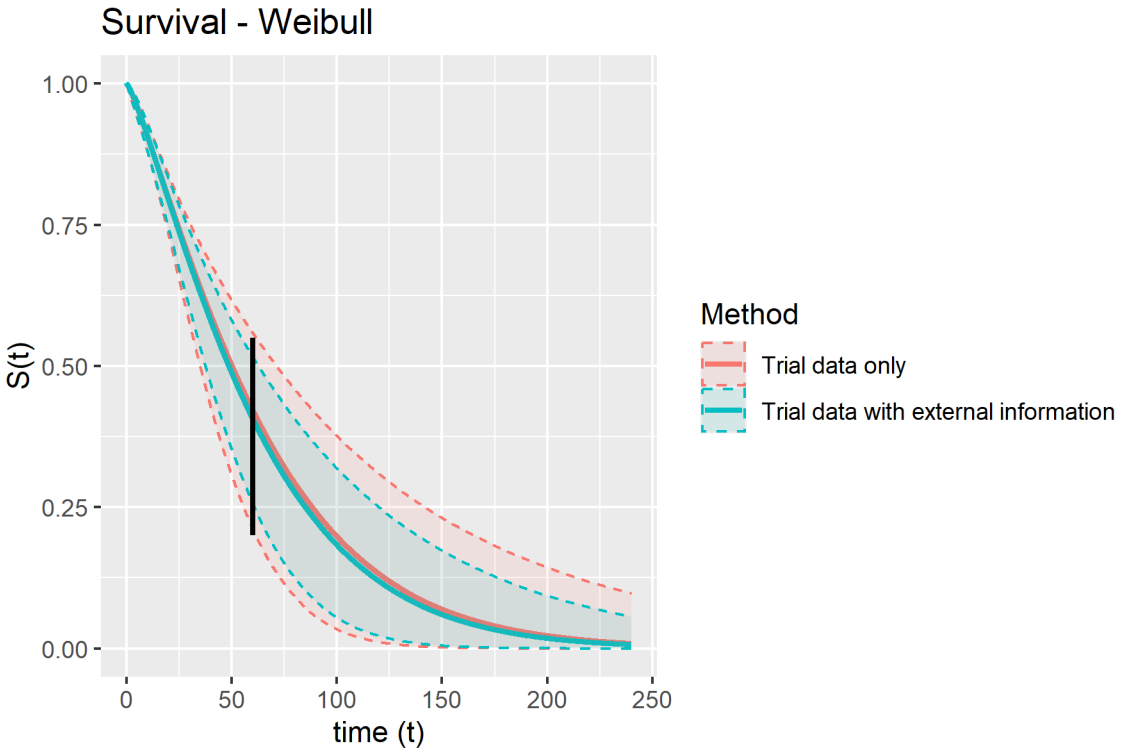

Parameters - Log-logistic
shape
scale
Corr: -0.842***
Trial data only: -0.852***
al data with external information: -0.796
shape
scale
3
2
1
0
4.5
4.0
3.5
0.00
0.25
0.50
0.75
3.5
4.0
4.5

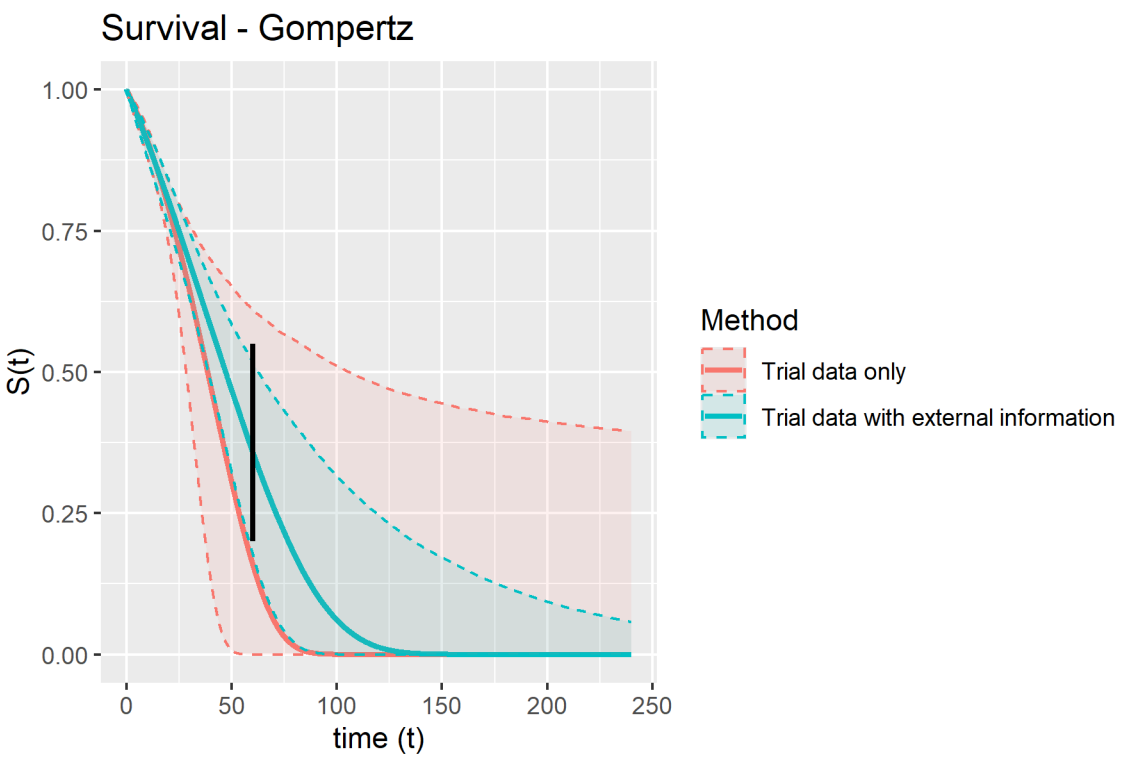


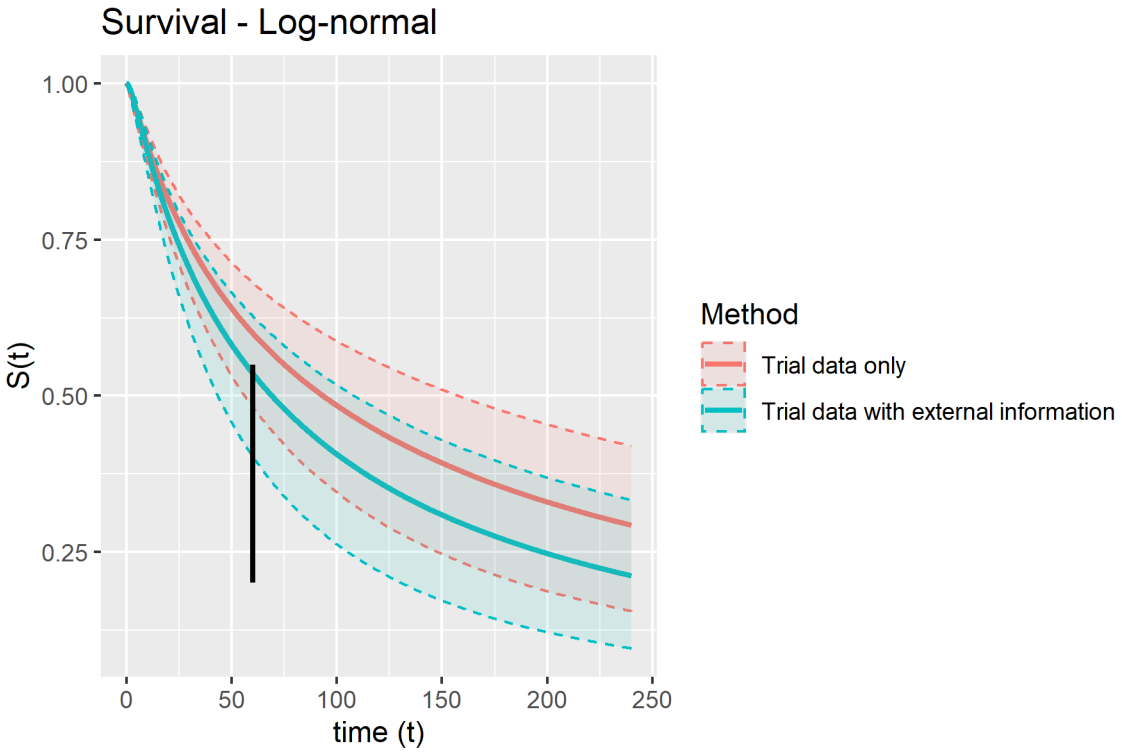
1.4 Survival comparison plots

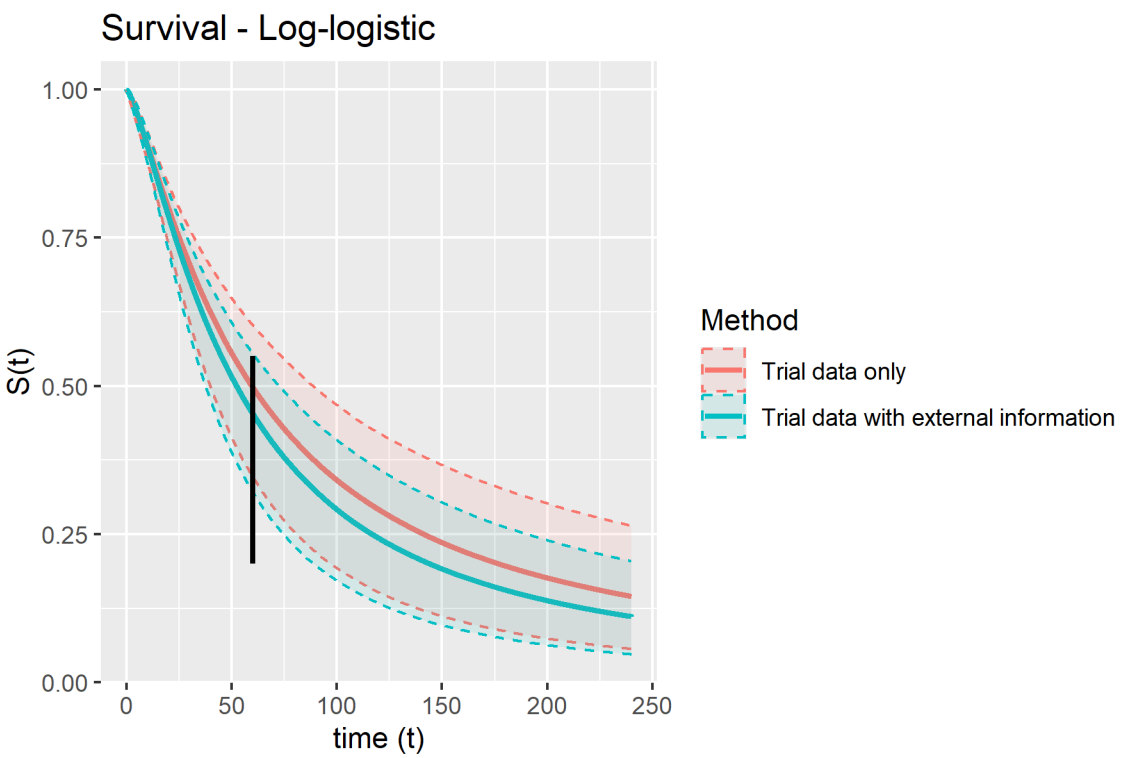
The below plots show the estimated survival curves (median and 95% confidence intervals) obtained using trial data only and combining trial data with external information, for the survival distributions under consideration. Vertical bars indicate the 95% probability interval for 5 year OS based on external information.

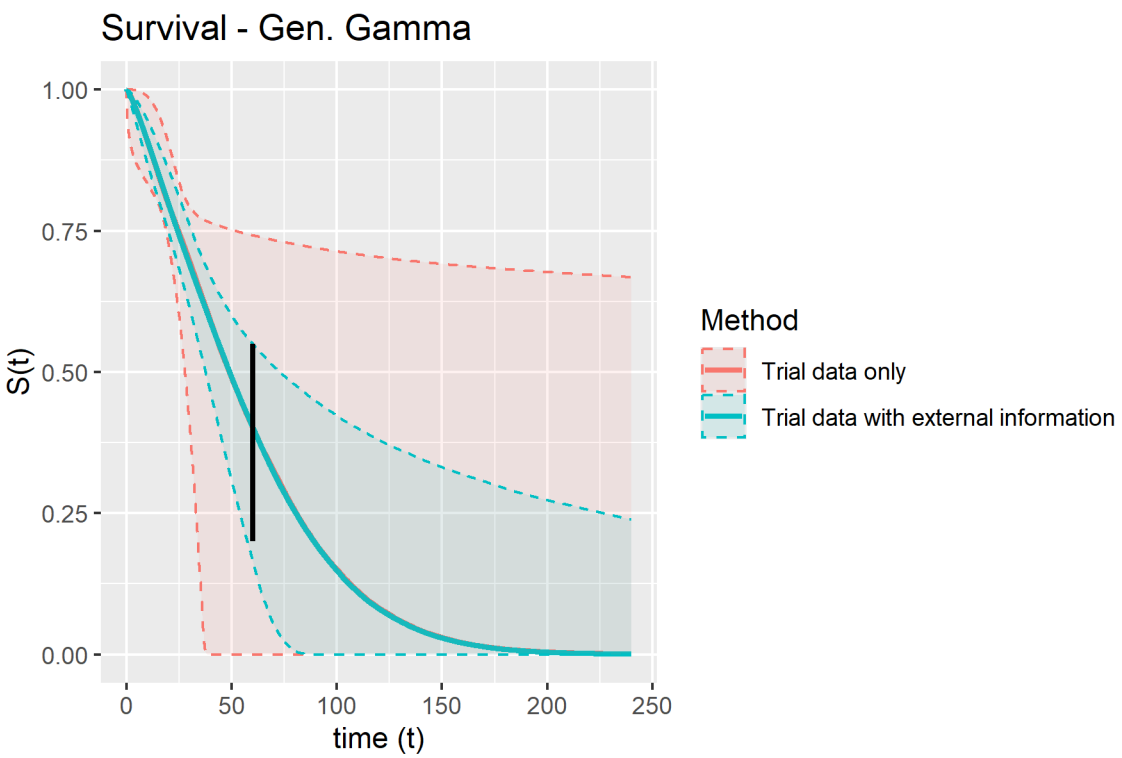



Survival - Weibull
1.00
0.75
0.50
0.25
0.00
S(t)
0
50
100
150
200
250
time (t)
Method
Trial data only
Trial data with external information


Survival - Gompertz
1.00
0.75
0.50
0.25
0.00
S(t)
0
50
100
150
200
250
time (t)
Method
Trial data only
Trial data with external information


Survival - Log-normal
1.00
0.75
0.50
0.25
S(t)
0
50
100
150
200
250
time (t)
Method
Trial data only
Trial data with external information


Survival - Log-logistic
1.00
0.75
0.50
0.25
0.00
S(t)
0
50
100
150
200
250
time (t)
Method
Trial data only
Trial data with external information


Survival - Gen. Gamma
1.00
0.75
0.50
0.25
0.00
S(t)
0
50
100
150
200
250
time (t)
Method
Trial data only
Trial data with external information

1.5 Parameter variability

Table 13 shows 'Generalised variance,' i.e., determinants of variance-covariance matrices, of the parameter estimates obtained using trial data only (MLE) and combining trial data withe external information (IS). Formally, the generalised variance gives the area of the 95% highest density ellipse and can be interpreted as a 1-parameter measure of parameter uncertainty.

Table 13: Generalised variance of model parameter distributions. Comparison between parameter estimates obtained using trial data only (MLE) and combining trial data with external information (IS).

Distribution	Distance	MLE.Variance	IS.Variance	Var..Ratio
exponential	0.0957855	0.0169492	0.0138444	0.8168192
weibull	0.0413797	0.0001670	0.0000827	0.4948562
gompertz	0.1474560	0.0000114	0.0000013	0.1140736
lognormal	0.3260830	0.0001878	0.0001435	0.7641277
llogis	0.1461999	0.0001754	0.0001134	0.6466361
gengamma	0.0546778	0.0003562	0.0000405	0.1138456

2 Comparison with expertsurv output

In this section we compare the outputs obtained from the importance sampling method described in the paper to that obtained using the Cooney & White method implemented in the expertsurv package. For the gen. Gamma model estimated using the expertsurv package, suitable Monte Carlo performance could not be obtained and hence results are not reported.

2.1 Plots of survival curves over time

Figures 1 to 5 show comparisons of survival curves (median and 95% confidence/credible intervals) obtained from the importance sampling method described in the paper, with those obtained from the fully Bayesian approach of Cooney & White implemented in the expertsurv package.

Survival curves for the exponential, Weibull, Gompertz and log-logistic models were almost identical between the two methods. For the log-normal model (Figure 4) there was a small but notable shift in the parameter distribution and corresponding survival estimate. The reason for this is not immediately clear. However, in both cases the impact was small and mostly evident in the tail of the survival distribution.

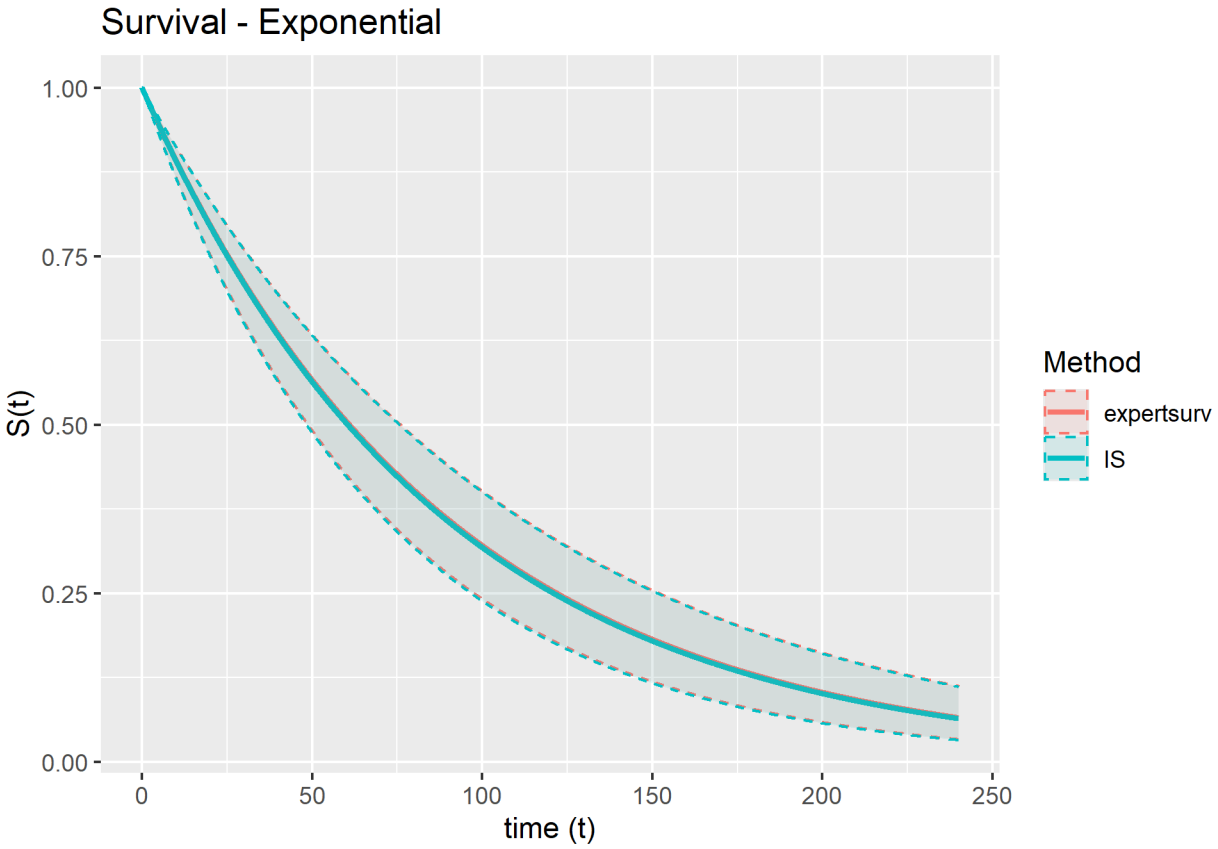


Figure 1: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the importance sampling (IS) and expertsurv methods, Exponential distribution

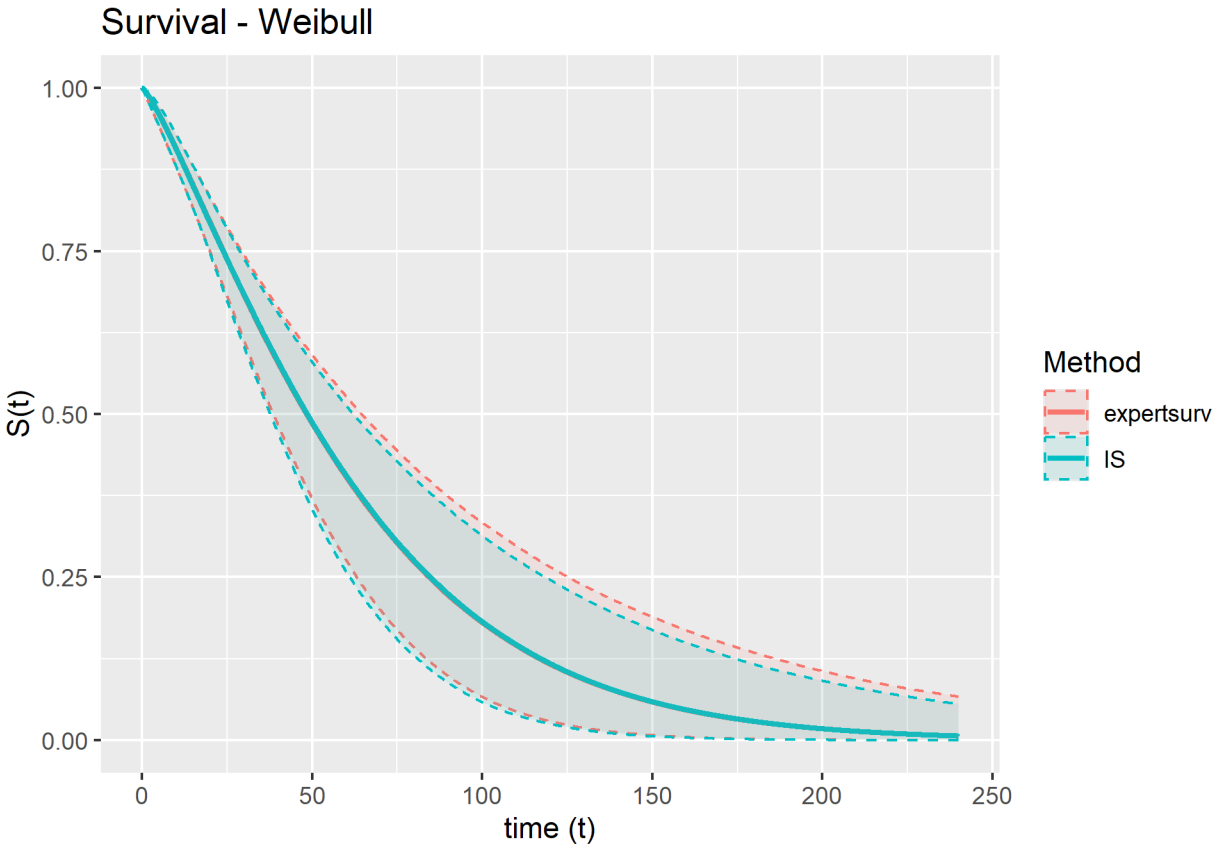


Figure 2: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the importance sampling (IS) and expertsurv methods, Weibull distribution

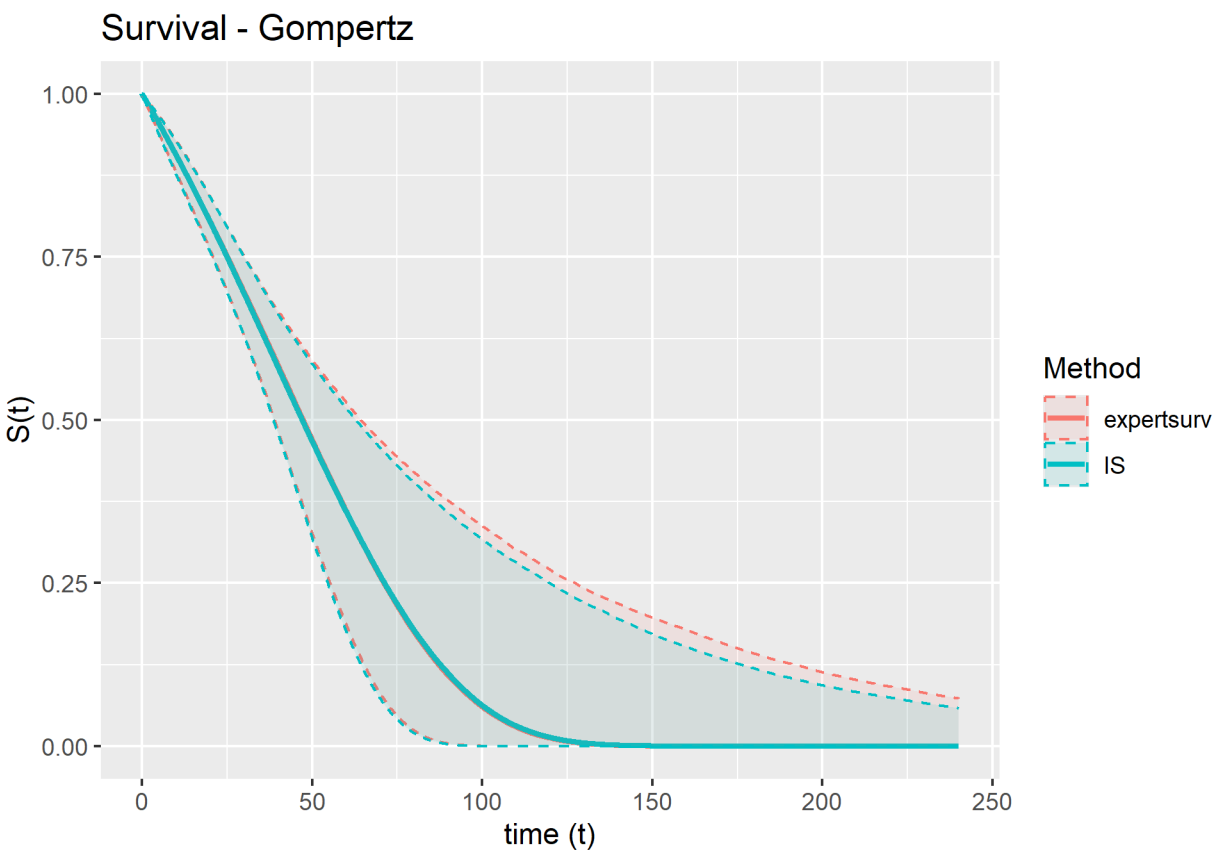


Figure 3: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the importance sampling (IS) and expertsurv methods, Log-normal distribution

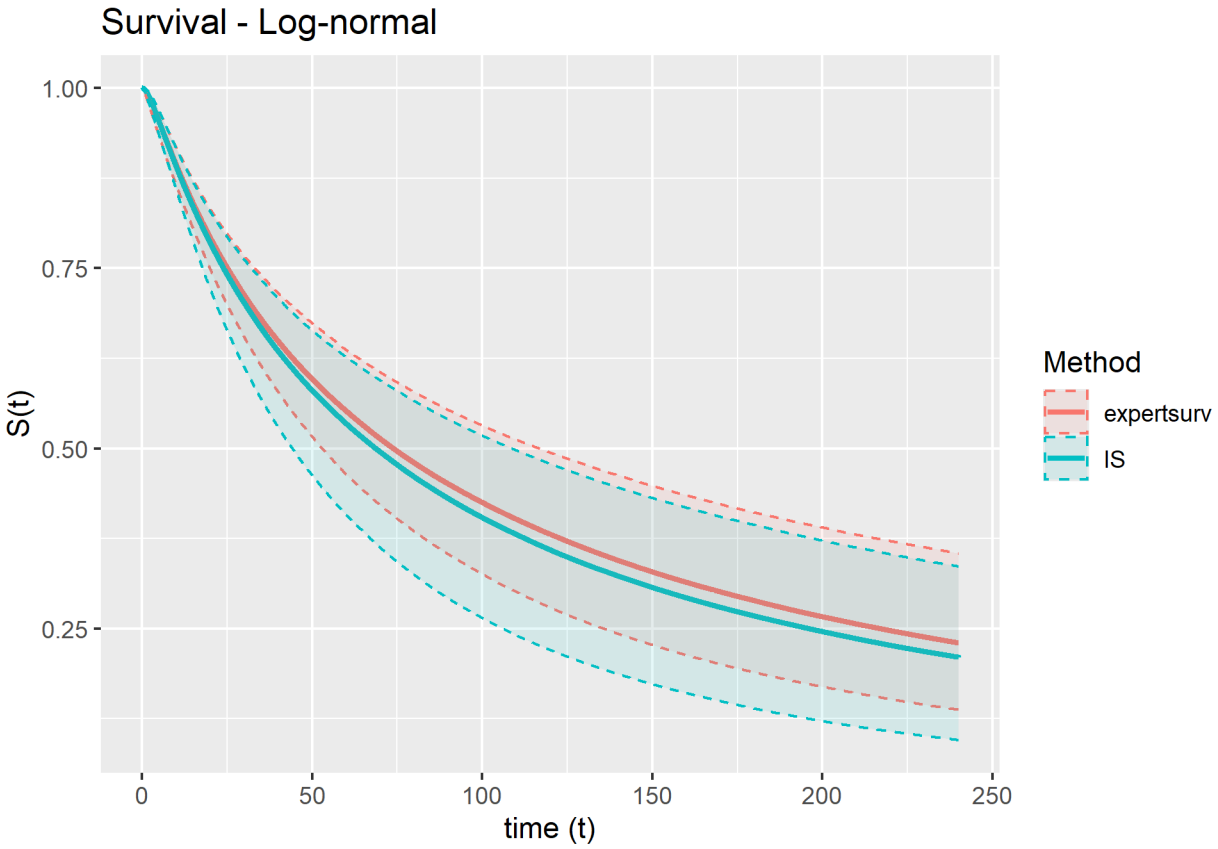


Figure 4: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the importance sampling (IS) and expertsurv methods, Log-logistic distribution

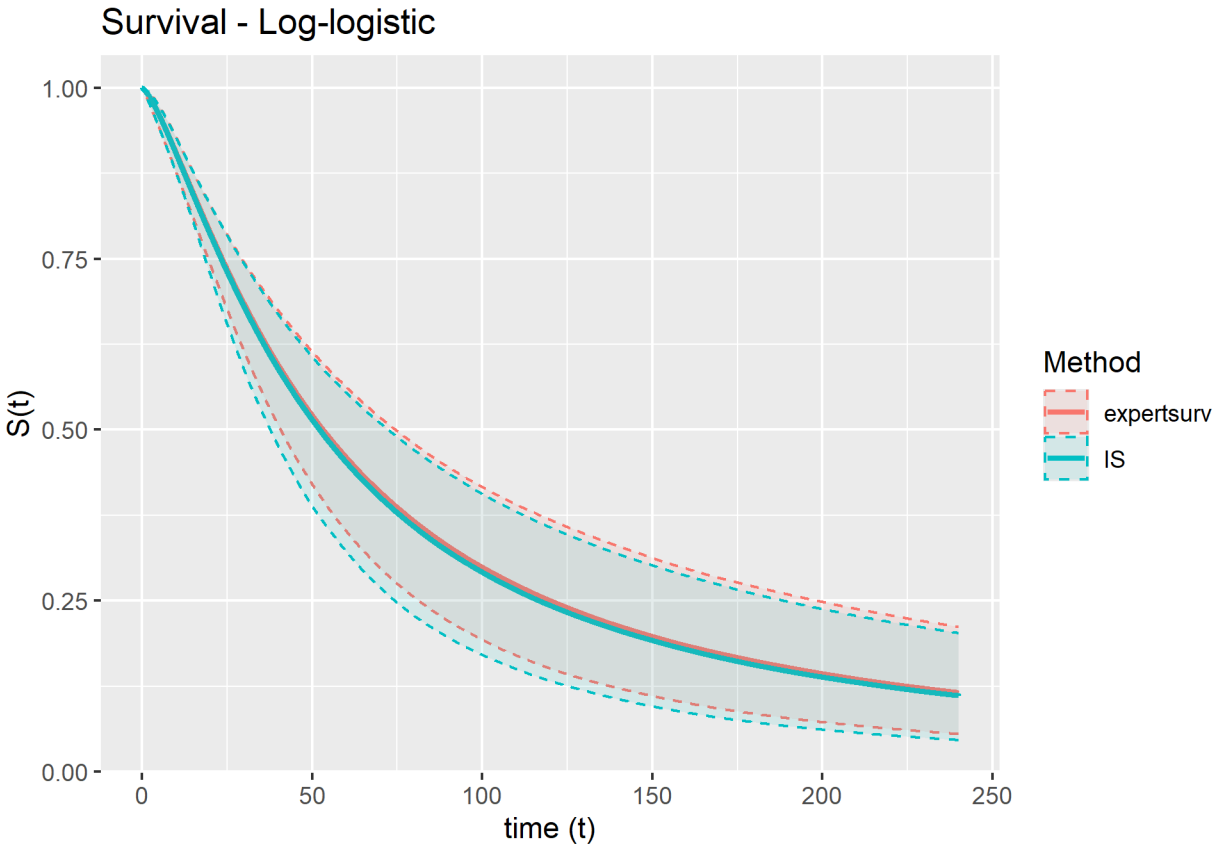


Figure 5: Survival curves (median and 95 percent confidence intervals) obtained by combining trial data with external information using the importance sampling (IS) and expertsurv methods, Gompertz distribution

2.2 Comparisons of parameter distributions

Figures 6 to 10 compare survival curve parameter distributions obtained from the importance sampling method described in the paper, with those obtained from the fully Bayesian approach of Cooney & Whilte implemented in the expertsurv package. Again, some differences are evident between the two methods for the log-normal model (Figure 9).

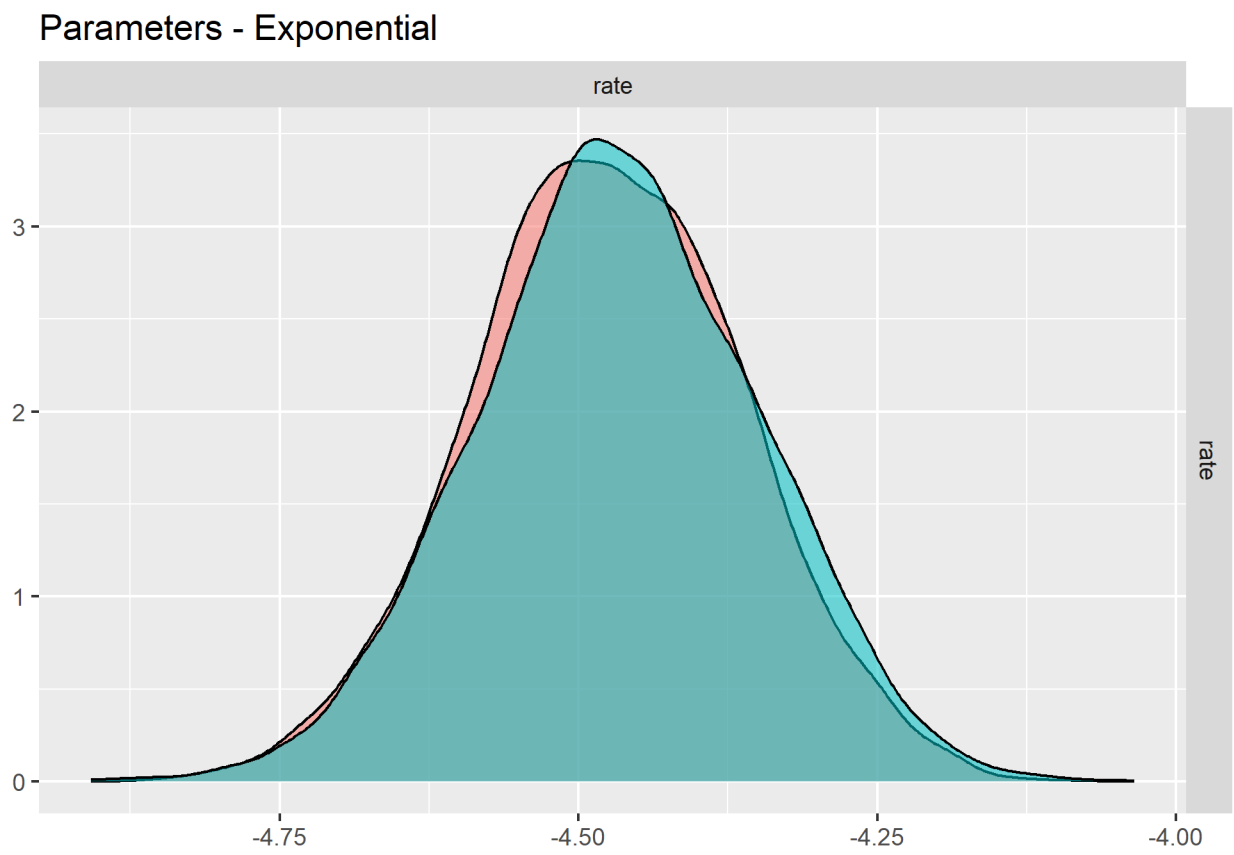


Figure 6: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the importance sampling (IS) and expertsurv methods, Exponential distribution

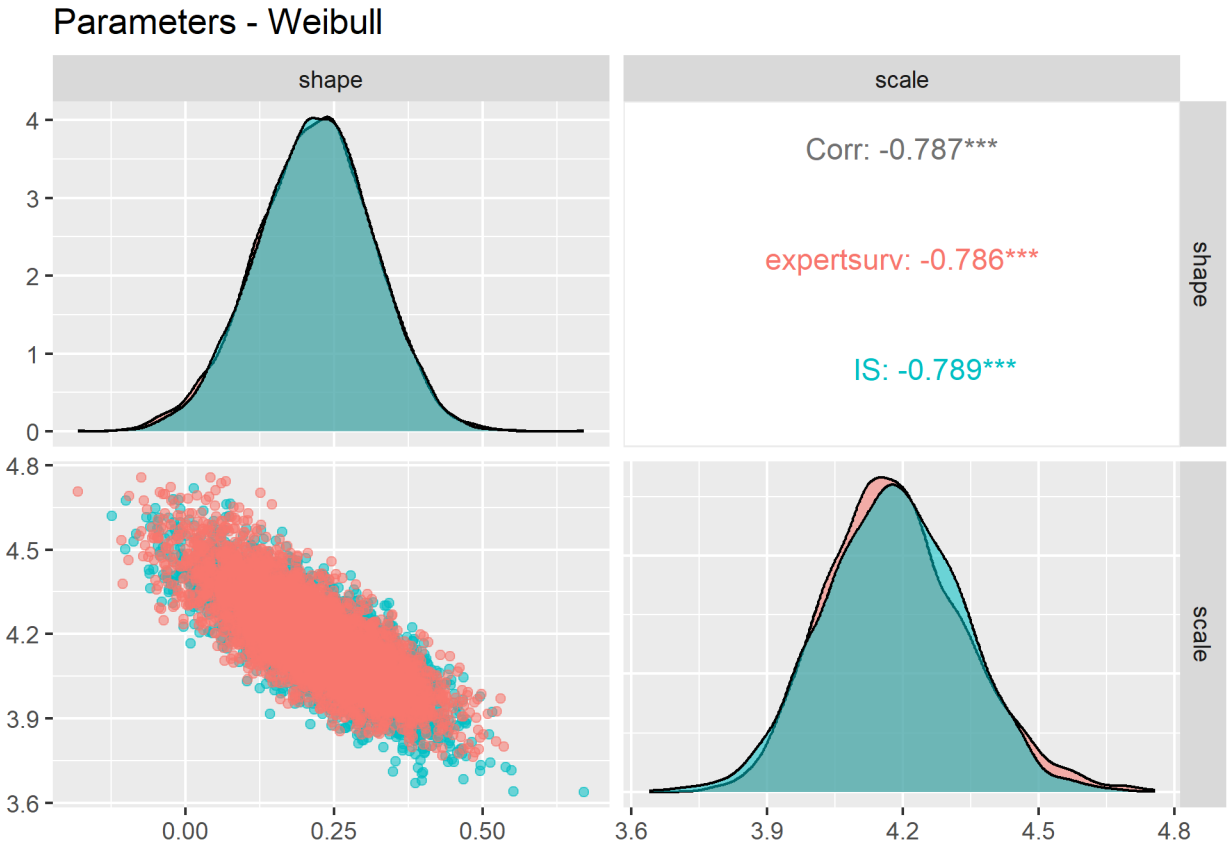


Figure 7: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the importance sampling (IS) and expertsurv methods, Weibull distribution

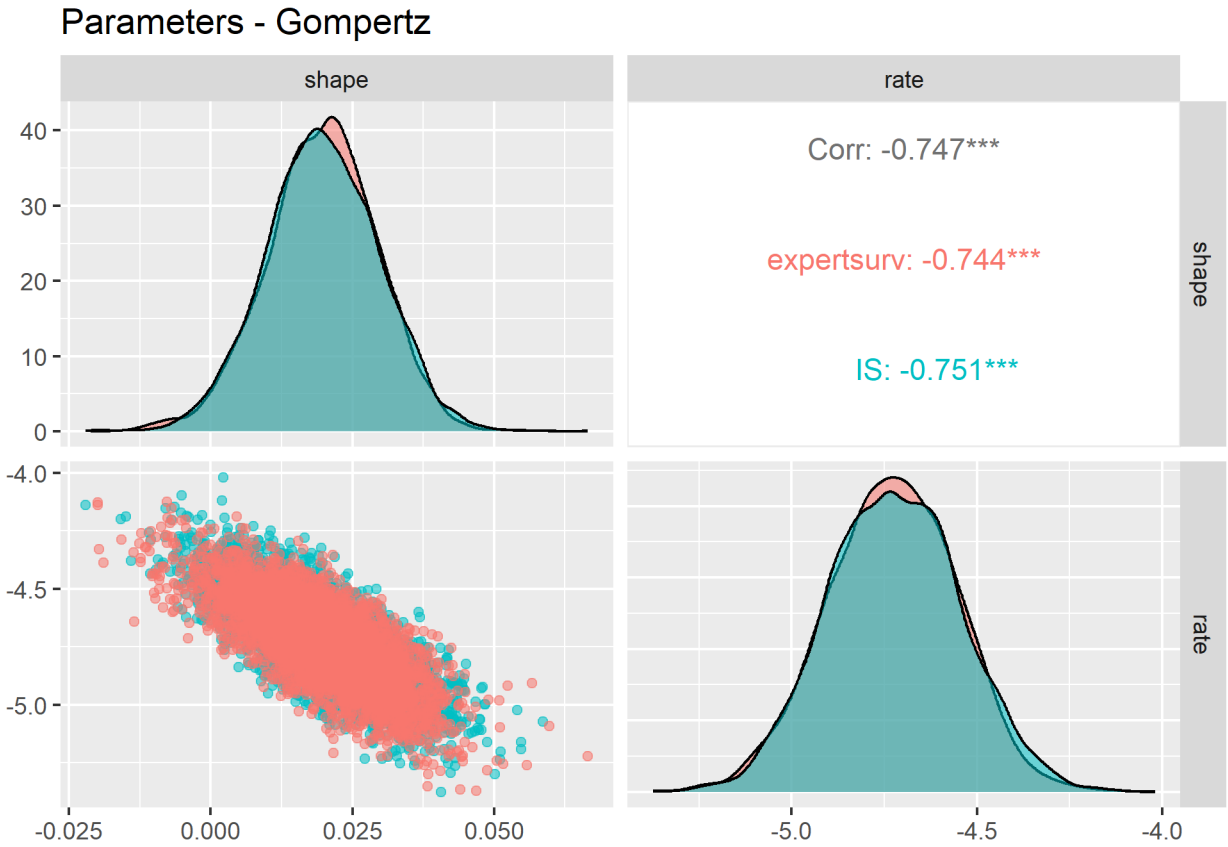


Figure 8: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the importance sampling (IS) and expertsurv methods, Log-normal distribution

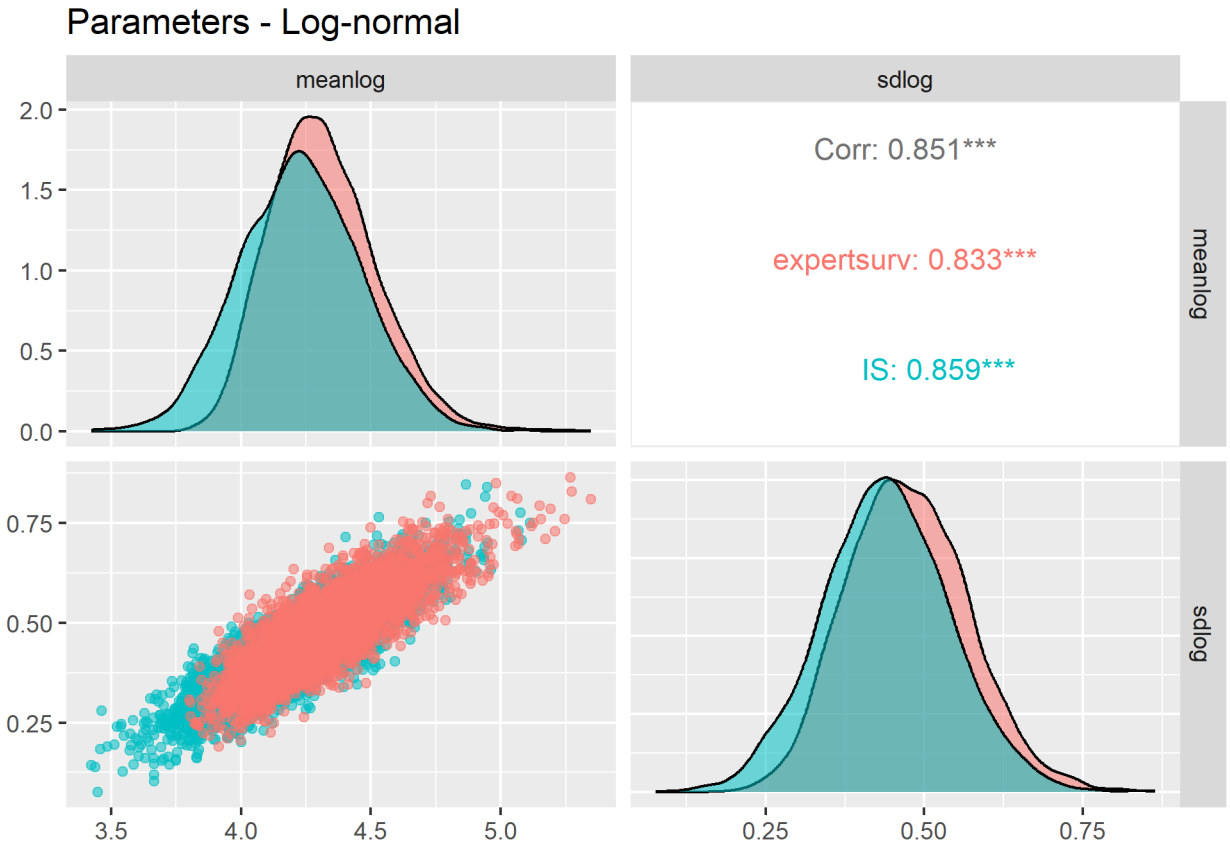


Figure 9: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the importance sampling (IS) and expertsurv methods, Log-logistic distribution

Table 14: Comparison of probabilistic total area under the curve (mean lifetime OS) estimates, obtained from combining trial data with external information using the importance sampling (IS) method described in the paper, and with the fully Bayesian method of Cooney and White implemented in the expertsurv package. Values shown are probabilstic mean and 95 percent confidence/credible intervals.

Distribution	expertsurv	IS	Mean.Diff	Var.Ratio
Exponential	88.48 (70.48, 109.73)	87.74 (69.78, 109.28)	-0.74	1.06
Gompertz	55.85 (39.56, 92.84)	53.89 (39.24, 85.24)	-1.96	0.44
Log-logistic	123.17 (77.65, 189.59)	119.06 (71.11, 181.96)	-4.11	1.01
Log-normal	202.54 (126.08, 311.84)	183.80 (97.93, 295.22)	-18.74	1.07
Weibull	62.43 (45.73, 90.16)	61.93 (44.18, 85.17)	-0.50	0.90

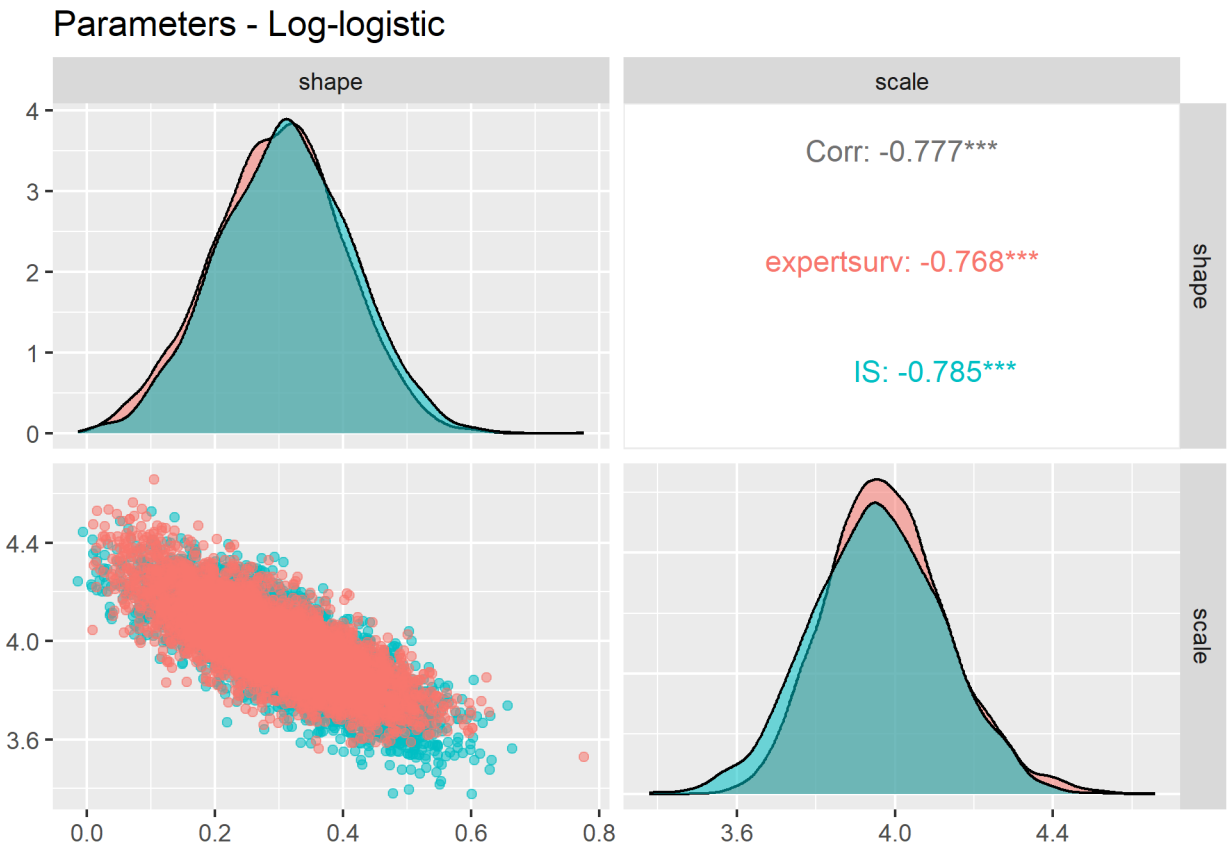


Figure 10: Survival curve parameter distributions and covariance estimates obtainedby combining trial data with external information using the importance sampling (IS) and expertsurv methods, Gompertz distribution

2.3 Comparisons of AUC distributions

Table 14 and Figure 11 compare probabilistic estimates of area under the curve between the two methods. In general, AUC measures (population) mean survival extrapolated over a lifetime, though in these examples survival is capped at 100 years since some curves plateau.

Table 15: Comparison of 5-year OS estimates, obtained from combining trial data with external information using the importance sampling (IS) method described in the paper, with the fully Bayesian method of Cooney and White implemented in the expertsurv package.

Distribution	Output	mean	median	sd	lwr.95	upr.95
Exponential	IS	50.16%	50.28%	4.03%	42.32%	57.75%
Exponential	expertsurv	50.47%	50.56%	3.87%	42.68%	57.88%
Gompertz	IS	35.63%	35.98%	8.87%	17.68%	51.87%
Gompertz	expertsurv	35.89%	36.01%	8.71%	18.84%	52.77%
Log-logistic	IS	44.89%	45.25%	5.98%	32.21%	55.46%
Log-logistic	expertsurv	45.88%	45.95%	5.37%	35.19%	56.30%
Log-normal	IS	52.94%	53.44%	5.72%	40.77%	62.69%
Log-normal	expertsurv	55.11%	55.20%	4.50%	46.42%	63.67%
Weibull	IS	40.05%	40.56%	6.65%	25.86%	51.30%
Weibull	expertsurv	40.32%	40.41%	6.53%	27.64%	52.69%

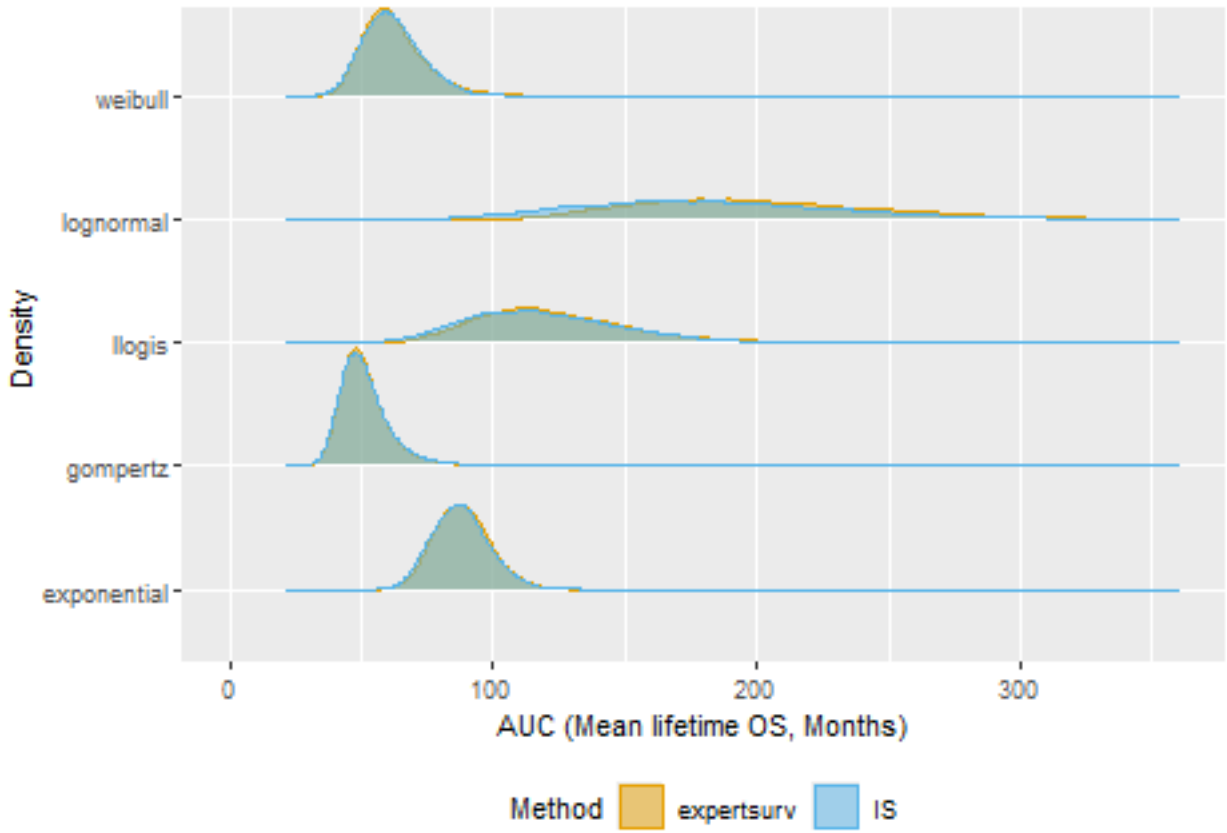


Figure 11: Density plots comparing probabilistic total area under the curve (mean OS) distributions obtained from combining trial data with external information using the importance sampling (IS) method described in the paper, with those obtained from the fully Bayesian method of Cooney and White implemented in the expertsurv package.

2.4 Comparisons of 5-year OS

Table 15 and Figure 12 compare probabilistic estimates of 5-year OS obtained from the two methods.

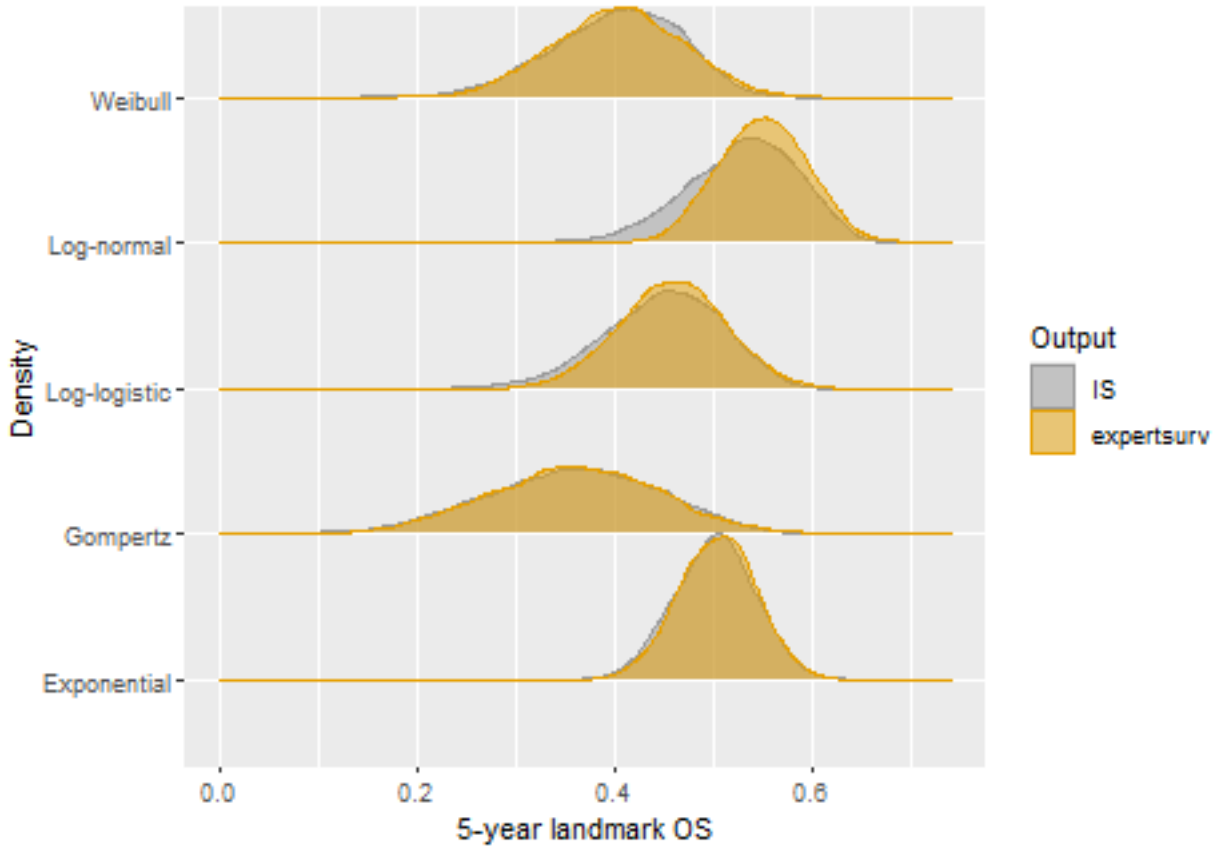


Figure 12: Density plots comparing 5-year OS estimates, obtained from combining trial data with external information using the importance sampling (IS) method described in the paper, with those obtained from the fully Bayesian method of Cooney and White implemented in the expertsurv package.